

GRANT NEILSON

Engineering focused Computational Biologist with seven years of industry experience building scalable and robust analysis pipelines, developing internal Python & R packages and processing large-scale whole genome and single cell sequencing datasets (500GB+). Committed to writing production-quality code through rigorous testing, version control, and thorough documentation. Active open-source contributor, passionate about developing reliable scientific software and advancing reproducible research through community-driven collaboration.

View this CV [online](#)

CONTACT

✉ grant.neilson5@gmail.com

🐙 [GitHub](#)

in [LinkedIn](#)

INDUSTRY EXPERIENCE

2026
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2023

Senior Computational Biologist

[CoSyne Therapeutics](#)

📍 London, UK

- Architected and deployed containerized bioinformatics infrastructure using Docker and AWS ECR, establishing scalable cloud-native solutions for multi-omic data analysis. Designed and implemented CI/CD pipelines for reproducible computational workflows, reducing deployment time by 60%.
- Lead developer and maintainer of multiple internal Python and R packages, establishing comprehensive testing frameworks, documentation standards, and code review processes. Implemented project templating system using Cruft, standardizing development practices across the organization.
- Engineered hyperscale cloud infrastructure on AWS for processing ~3 million single-cell sequencing samples, implementing automated alignment and QC pipelines. Optimized compute resource allocation and parallelization strategies, achieving 50% cost reduction and 30% performance improvement.
- Contributed to open-source NF-core Sarek pipeline (v3.4.1), implementing performance optimizations and workflow improvements. Developed custom Nextflow modules for variant calling and quality control processes.
- Established software engineering best practices including version control workflows, code review standards, unit testing frameworks, and comprehensive documentation. Mentored team members on software development methodologies and DevOps practices.

2023
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2020

Computational Biologist

[Fios Genomics](#)

📍 Edinburgh, UK

- Directed projects focusing on high-dimensional omics data types (WES, WGS, scRNA-seq, proteomics), delivering comprehensive analyses and reports for clients.
- Developed in-depth knowledge of statistical methodologies, mastering model selection and fitting techniques, including SVM, elastic net, random forest, and linear/logistic regression, while accounting for covariates.
- Conducted complex regression model analyses for biomarker discovery in bladder and lung cancer.
- Actively contributed to internal R&D by developing and maintaining specialized R packages, enhancing the team's analytical capabilities.
- Led client calls, effectively communicating complex analytical results, ensuring clarity and client understanding.

LANGUAGES

📊 R
🐍 Python
🔗 Bash
🔗 Nextflow

TECHNOLOGIES

🐙 GitLab/GitHub
🐳 Docker
aws aws
🖥 Azure

OPEN SOURCE & PUBLIC PROJECTS

🐙 [Sarek](#)
🐙 [bixverse](#)
🐙 [mainfoldsR](#)

Made with the R package [pagedown](#).

The source code is available on [GitHub](#).

Last updated on 2026-02-18.

2020
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2019

Computational Biologist

Complex Diseases Epigenetics group, Exeter Medical School

📍 Exeter, UK

- Developed novel bioinformatics pipelines for Epigenome/Genome wide association studies, mQTL analysis and quality control of Illumina EPIC array data.
- Took the lead on a project performing EWAS and GWAS analysis in ADHD patients. This involved fitting linear and logistic regression models while accounting for covariates.
- Was selected for a secondment to New York to work at the Mount Sinai medical school as a visiting bioinformatician. This was unfortunately cancelled due to COVID-19.



EDUCATION

2019
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2015

MSCi Biology

University of Southampton

📍 Southampton, UK

- Grade : First Class Honors
- Winner of the British Society for Immunology Graduate of the year award.
- Recipient of the summer studentship from the Genetics Society (2017).

2015
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2013

School

Merchiston Castle School

📍 Edinburgh, UK

- A Levels: Biology, Chemistry, Maths



SELECTED PUBLICATIONS

2025

Multi modal single cell foundation models via dynamic token adaptation

BioRxiv

- Wenmin Zhao, AnaSolaguren-Beascoa, Grant Neilson, Regina Reynolds, Louwai Muhammed, Liisi Laaniste, Sera Aylin Cakiroglu
- Role: Data Preprocessing
- doi: <https://doi.org/10.1101/2025.04.17.649387>

2025

Detecting cell level transcriptomic changes of Perturb seq using Contrastive Fine tuning of Single Cell Foundation Models

BioRxiv

- Wenmin Zhao, AnaSolaguren-Beascoa, Grant Neilson, Regina Reynolds, Louwai Muhammed, Liisi Laaniste, Sera Aylin Cakiroglu
- Role: Data Preprocessing
- doi: <https://doi.org/10.1101/2025.04.17.649395>